

CSE4203: Computer Graphics
Lecture – 8 (Part - B)
Fractal Geometry

Outline

- L-systems

L-systems

- Lindenmayer-systems or L-system is developed by 1968 by biologist Aristid Lindenmayer
- It's a grammar based technique
 - Represent shape as string of symbol
 - Each symbol has meaning in drawing shape
- Two parts:
 - Grammar for generating strings
 - Rendering algorithm for interpreting strings as shapes

L-system structure

- L-systems are commonly known as parametric L systems, defined as a tuple

$$\mathbf{G} = (V, \omega, P)$$

- **V (the alphabet)** is a set of symbols containing both elements that can be replaced (variables) and those which cannot be replaced ("constants" or "terminals")
- **ω (start, axiom or initiator)** is a string of symbols from V defining the initial state of the system
- **P is a set of production rules** or productions defining the way variables can be replaced with combinations of constants and other variables.

Example - 1: algae (1/4)

- L-system for modelling the growth of algae
variables : **A B**
constants : **none**
axiom : **A**
rules : **(A $\rightarrow$ AB), (B $\rightarrow$ A)**

Example - 1: algae (2/4)

$n = 0 : A$

$n = 1 : AB$

$n = 2 : ABA$

$n = 3 : ABAAB$

$n = 4 : ABAABABA$

$n = 5 : ABAABABAABAAB$

$n = 6 : ABAABABAABAABAABAABA$

$n = 7 : ABAABABAABAABAABAABAABAABAABAABAABAABA$

axiom : A

rules : $(A \rightarrow AB), (B \rightarrow A)$

Example - 1: algae (3/4)

axiom : A
rules : (A $\rightarrow$ AB), (B $\rightarrow$ A)

n=0: A start (axiom/initiator)

 / \

n=1: A B the initial single A spawned into AB by rule (A $\rightarrow$ AB), rule (B $\rightarrow$ A)

couldn't be applied

 / | \

n=2: A B A former string AB with all rules applied, A spawned into AB again,

former B turned into A

 / | | | \

n=3: A B A A B note all A's producing a copy of themselves in the first place, then

a B, which turns ...

 / | | | \ | \ \

n=4: A B A A B A B A ... into an A one generation later, starting to spawn/repeat/recurse

then

Example - 1: algae (4/4)

axiom : A
rules : (A → AB), (B → A)

n=0: A start (axiom/initiator)

 / \

n=1: A B the initial single A spawned into AB by rule (A → AB), rule (B → A)

couldn't be applied

 / | \

n=2: A B A former string AB with all rules applied, A spawned into AB again,

former B turned into A

 / | | | \

n=3: A B A A B note all A's producing a copy of themselves in the first place, then

a B, which turns ...

 / | | | | \ \

n=4: A B A A B A B A ... into an A one generation later, starting to spawn/repeat/recurse

If we count the length of each string, we obtain the Fibonacci sequence of numbers: 1 2 3 5 8 13 21 34 55 89 ...

Example - 2: Cantor set (1/6)

variables : **A B**

constants : **none**

axiom : **A**

rules : **(A $\rightarrow$ ABA), (B $\rightarrow$ BBB)**

A: Draw a line forward

B: Move forward without drawing

Example - 2: Cantor set (2/6)

n = 0: A

n = 1: ABA

n = 2: ABABBBABA

n = 3: ABABBBABABBBBBBBBBBABABBBABA

variables : A B

constants : none

axiom : A

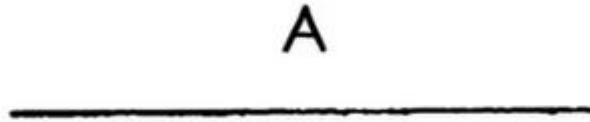
rules : (A $\rightarrow$ ABA), (B $\rightarrow$ BBB)

A: Draw a line forward

B: Move forward without drawing

Example - 2: Cantor set (3/6)

$n = 0$: A



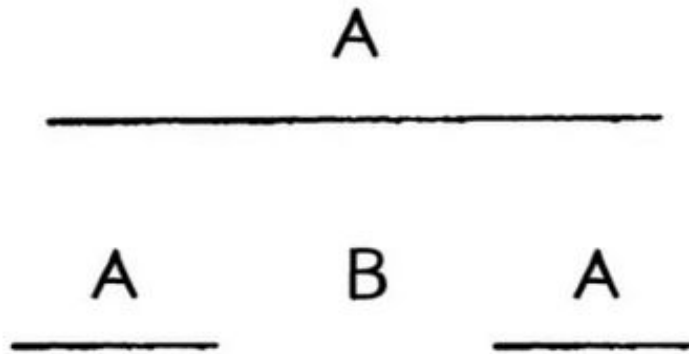
A: Draw a line forward

B: Move forward without drawing

Example - 2: Cantor set (4/6)

n = 1: ABA

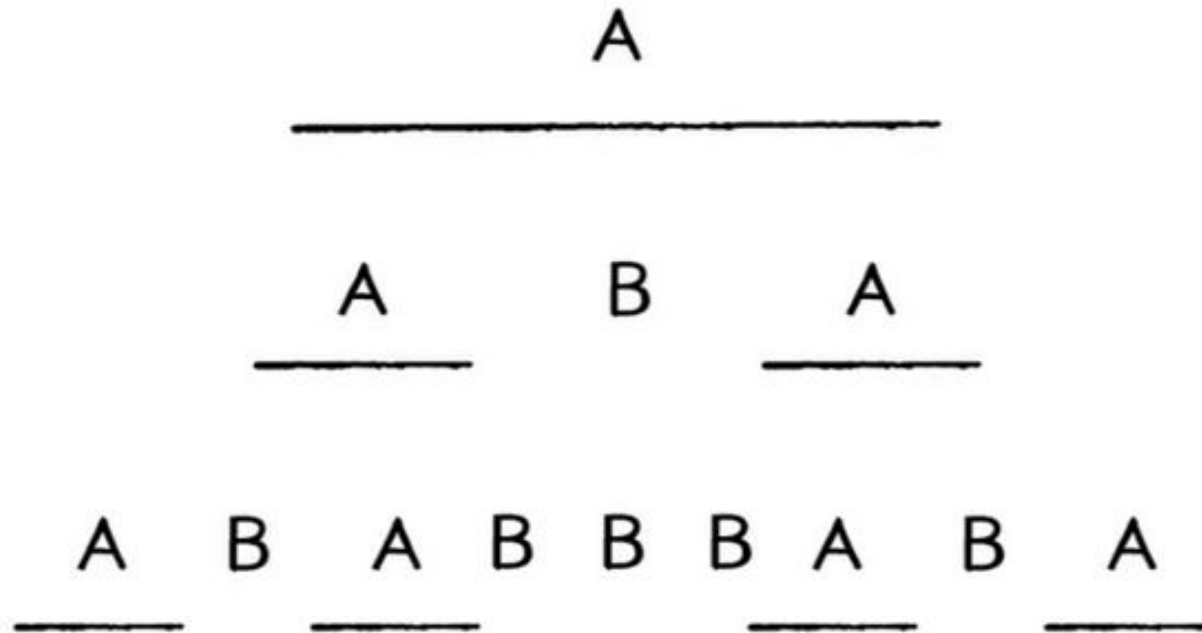
A: Draw a line forward
B: Move forward without drawing



Example - 2: Cantor set (5/6)

n = 2: ABABBBABA

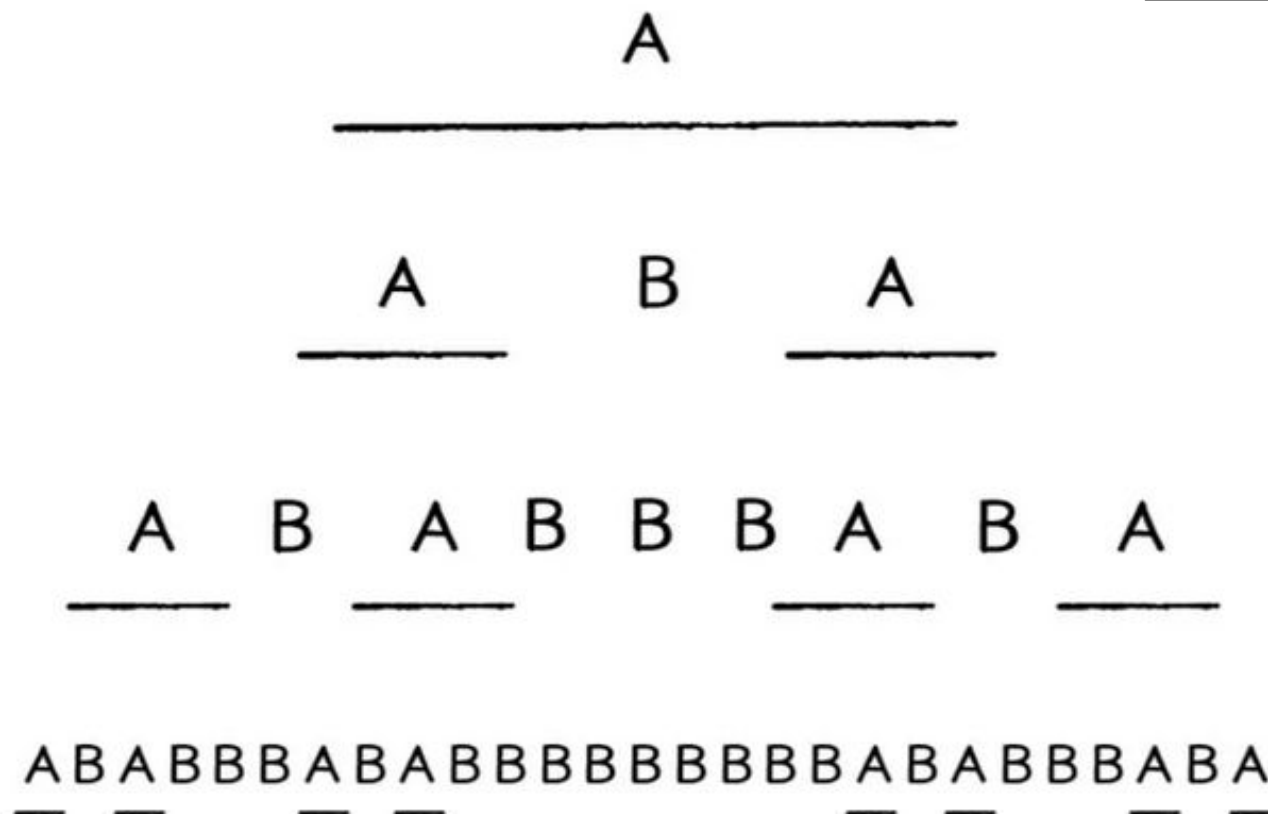
A: Draw a line forward
B: Move forward without drawing



Example - 2: Cantor set (6/6)

n = 3: ABABBBBABABBBBBBBBBBABABBBABA

A: Draw a line forward
B: Move forward without drawing



Example - 3: Koch Snowflake (1/5)

variables : **F**

constants : **+**, **-**

axiom : **F**

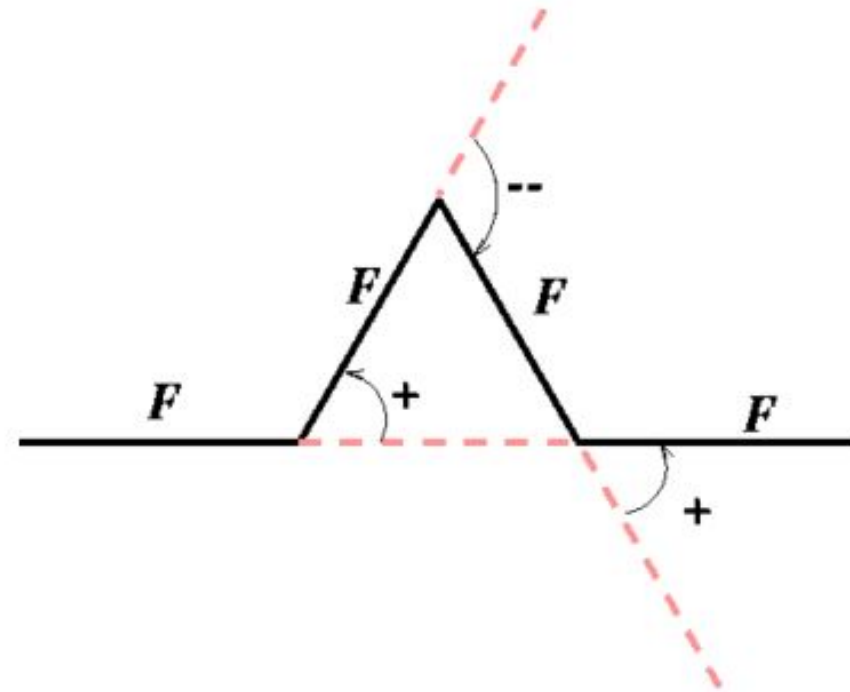
rules : **F** $\rightarrow$ **F+F--F+F**

Angle: 60 degrees

F: forward one unit

+ : turn left 60 degrees

- : turn right 60 degrees



Example - 3: Koch Snowflake (2/5)

n = 0: F

n = 1: F+F--F+F

n = 2: F+F--F+FF+F--F+FF+F--F+FF+F--F+F

variables : F, +, -

constants : none

axiom : F

rules : F → F+F--F+F

Angle: 60 degrees

F: forward one unit

+ : turn left 60 degrees

- : turn right 60 degrees

Example - 3: Koch Snowflake (3/5)

n = 0: F

F: forward one unit
+ : turn left 60 degrees
- : turn right 60 degrees



Example - 3: Koch Snowflake (4/5)

$n = 1$: F+F--F+F

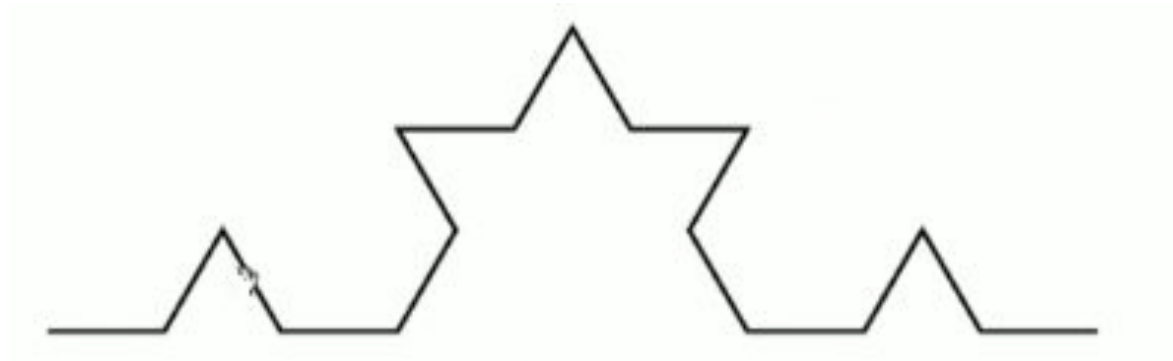
F: forward one unit
+: turn left 60 degrees
-: turn right 60 degrees



Example - 3: Koch Snowflake (5/5)

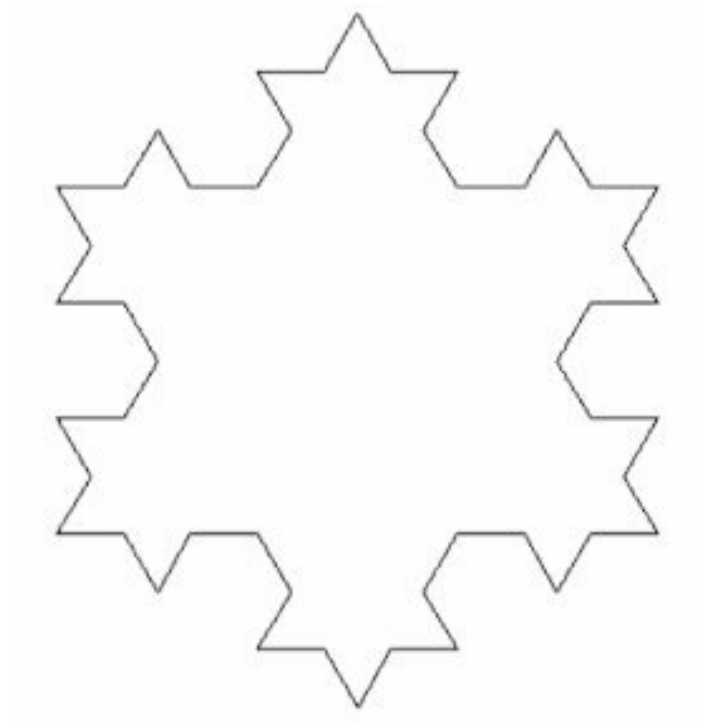
n = 2: F+F--F+F+F+F--F+F--F+F--F+F+F+F--F+F

F: forward one unit
+ : turn left 60 degrees
- : turn right 60 degrees



Example - 3: Koch Snowflake (5/5)

For all 3 sides of triangle



Example - 4: Fractal (binary) tree (1/10)

variables : **0 1**

constants : **[]**

axiom : **0**

rules : **(1 → 11), (0 → 1[0]0)**

Angle: 45 degrees

0: draw a line segment ending in a leaf

1: draw a line segment

[: push position and angle, turn left 45 degrees

]: pop position and angle, turn right 45 degrees

Example - 4: Fractal (binary) tree (2/10)

n = 0: 0

n = 1: 1[0]0

n = 2: 11[1[0]0]1[0]0

n = 3: 1111[11[1[0]0]1[0]0]11[1[0]0]1[0]0

variables : 0 1

constants : []

axiom : 0

rules : (1 → 11), (0 → 1[0]0)

Angle: 45 degrees

0: draw a line segment ending in a leaf

1: draw a line segment

[: push position and angle, turn left 45 degrees

] : pop position and angle, turn right 45 degrees

Example - 4: Fractal (binary) tree (3/10)

n = 0: 0

0: draw a line segment ending in a leaf



Example - 4: Fractal (binary) tree (4/10)

n = 1: 1[0]0

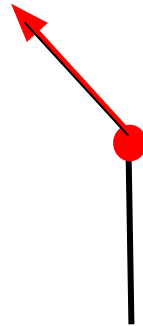
1: draw a line segment



Example - 4: Fractal (binary) tree (5/10)

n = 1: 1[0]0

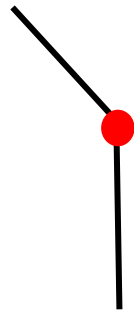
[: push position and angle, turn left 45 degrees



Example - 4: Fractal (binary) tree (6/10)

n = 1: 1[0]0

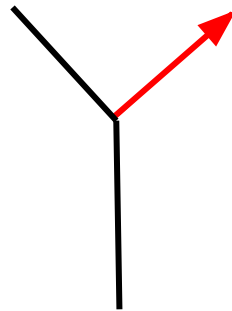
0: draw a line segment ending in a leaf



Example - 4: Fractal (binary) tree (7/10)

n = 1: 1[0]0

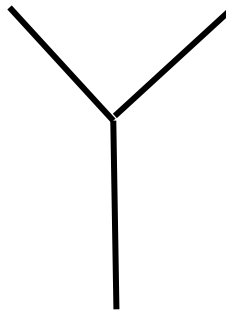
] : pop position and angle, turn right 45 degrees



Example - 4: Fractal (binary) tree (8/10)

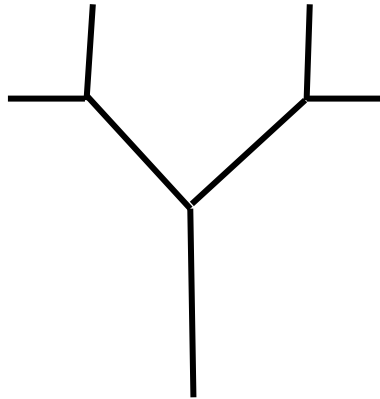
n = 1: 1[0]0

0: draw a line segment ending in a leaf



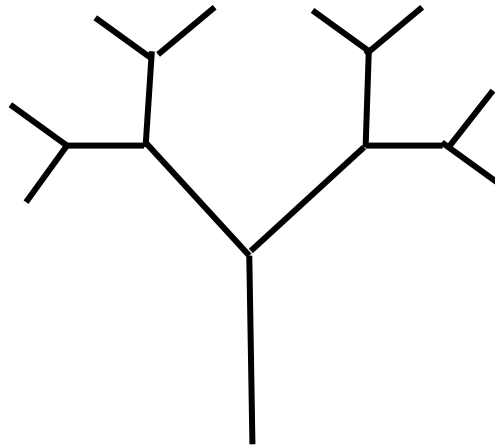
Example - 4: Fractal (binary) tree (9/10)

n = 2: 11[1[0]0]1[0]0



Example - 4: Fractal (binary) tree (10/10)

n = 3: 1111[11[1[0]0]1[0]0]11[1[0]0]1[0]0



Example - 5: Fractal Plant

variables : **X F**

constants : **+ - []**

axiom : **X**

rules : **(X → F+[[X]-X]-F[-FX]+X), (F → FF)**

Angle: 25 degrees

F: draw forward

X: do Nothing

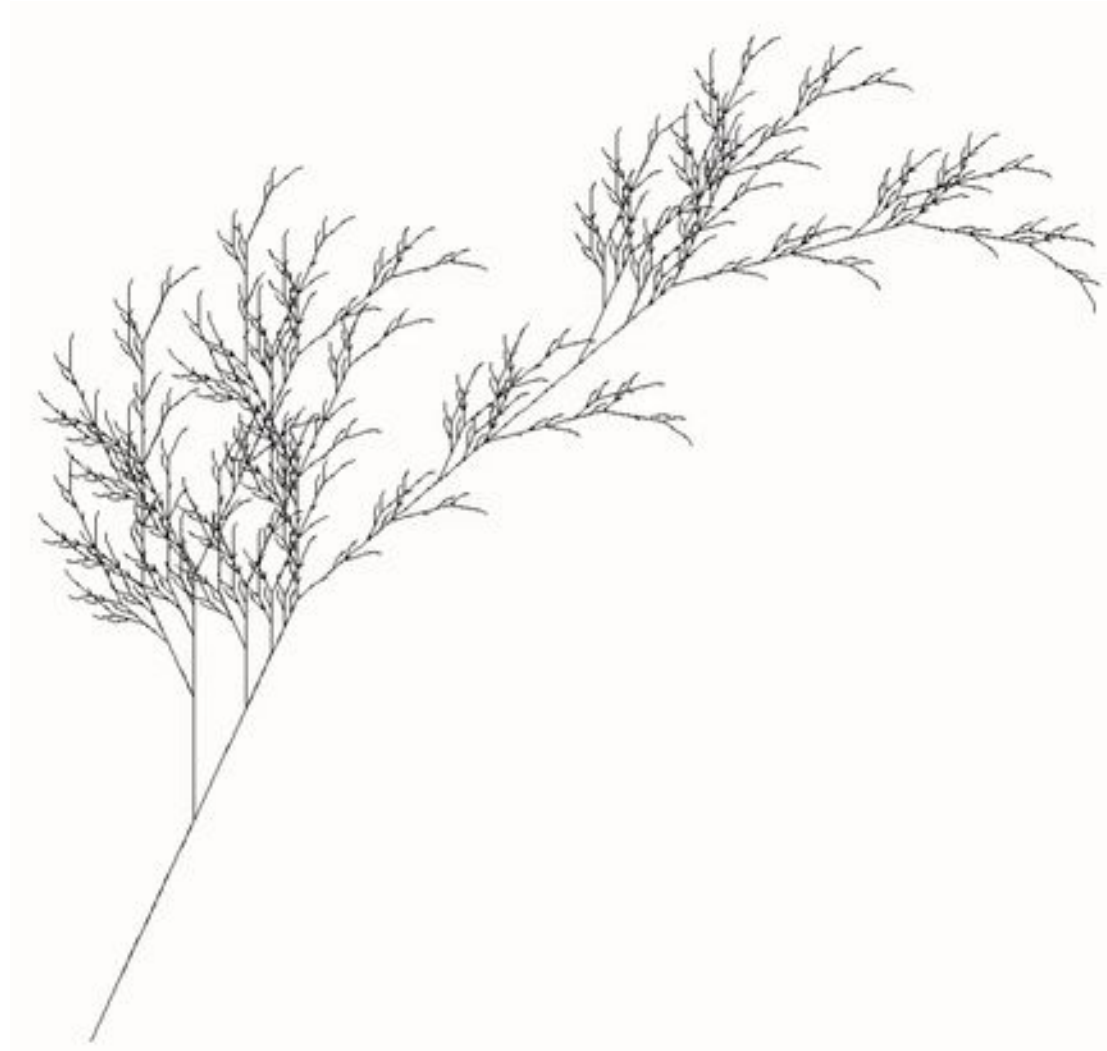
+ : turn left 25 degrees

- : turn right 25 degrees

[: push state

] : pop state

Example - 5: Fractal Plant



Fractal plant for $n = 6$ (GIF available [here](#))

Example - 5: Fractal Plant



a
 $n=5, \delta=25.7^\circ$
 F
 $F \rightarrow F[+F]F[-F]F$



b
 $n=5, \delta=20^\circ$
 F
 $F \rightarrow F[+F]F[-F][F]$



c
 $n=4, \delta=22.5^\circ$
 F
 $F \rightarrow FF - [-F+F+F] +$
 $[+F-F-F]$



d
 $n=7, \delta=20^\circ$
 X
 $X \rightarrow F[+X]F[-X]+X$
 $F \rightarrow FF$



e
 $n=7, \delta=25.7^\circ$
 X
 $X \rightarrow F[+X][-X]FX$
 $F \rightarrow FF$



f
 $n=5, \delta=22.5^\circ$
 X
 $X \rightarrow F - [[X]+X] + F[+FX] - X$
 $F \rightarrow FF$

Practice Problems

Apply the concept of L-systems for third iteration ($n = 3$) to draw -

- Sierpinski Triangle
- Koch curve
- Dragon curve

Further Reading

- <https://www.wikiwand.com/en/L-system>
- <http://paulbourke.net/fractals/lsys/>